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7.1.3 What you will learn

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This week is all about solving nonsingular linear systems with matrices that are sparse (have enough zero entries that it is worthwhile to exploit them).

Upon completion of this week, you should be able to

- Exploit sparsity when computing the Cholesky factorization and related triangular solves of a banded matrix.
- Derive the cost for a Cholesky factorization and related triangular solves of a banded matrix.
- Utilize nested dissection to reduce fill-in when computing the Cholesky factorization and related triangular solves of a sparse matrix.
- Connect sparsity patterns in a matrix to the graph that describes that sparsity pattern.
- Relate computations over discretized domains to the Jacobi, Gauss-Seidel, Successive Over-Relaxation (SOR) and Symmetric Successive Over-Relaxation (SSOR) iterations.
- Formulate the Jacobi, Gauss-Seidel, Successive Over-Relaxation (SOR) and Symmetric Successive Over-Relaxation (SSOR) iterations as splitting methods.
- Analyze the convergence of splitting methods.

Discussion

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